ETH Zürich
Department of Physics

The Berezinskii-Kosterlitz-Thouless Transition in the 2D XY Model

Lara Turgut **Francesco Conoscenti** **Ben Bullinger**

Computational Statistical Physics (402-0812-00L), Spring 2025

Outline

1. Introduction
2. Theoretical Background
3. Monte Carlo Methods
4. Implementation Details
5. Results and Analysis
6. Conclusion

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The 2D XY Model

- XY model: classical spins on a 2D lattice, constrained to a plane with $O(2)$ symmetry.

$$H = -J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) \tag{1}$$

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  - **Mermin-Wagner theorem:** No spontaneous breaking of continuous symmetries (hence no true long-range order) at $T > 0$ in $d \leq 2$.
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  - Yet, the 2D XY model *does* exhibit a phase transition!
- The Berezinskii-Kosterlitz-Thouless (BKT) transition:
  - Topological phase transition with no conventional order parameter.
  - Transition from quasi-long-range order (QLRO) to exponential disorder.
  - Driven by the unbinding of vortex-antivortex pairs.

Physical Significance

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  - Quantum spin systems
- Universal mechanism: A beautiful example of universality in critical phenomena

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Algebraic Order in the 2D XY Model

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$$g(r) = \langle e^{i(\theta(\mathbf{r}) - \theta(0))} \rangle \approx \left(\frac{r}{a} \right)^{-\eta(T)} \quad \text{with} \quad \eta(T) = \frac{k_B T}{2\pi J} \tag{3}$$

- Key property: Algebraic decay of correlations at low temperature.
  - **Low-T phase (QLRO):** $g(r) \sim r^{-\eta(T)}$ with temperature-dependent exponent.
  - **High-T phase (Disordered):** $g(r) \sim e^{-r/\xi(T)}$ (exponential decay).

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  - **High-T phase (Disordered):** $g(r) \sim e^{-r/\xi(T)}$ (exponential decay).
- This algebraic decay = **Quasi-Long-Range Order (QLRO)**.
  - Infinite correlation length $\xi = \infty$ in the low-T phase.
  - System is critical at all temperatures below T_{KT}.

The Role of Vortices

- Vortices: topological defects with winding of $2\pi n$ in phase field

$$\oint \nabla\theta \cdot d\mathbf{l} = 2\pi n \tag{4}$$

- Energy of a single vortex (with $n = 1$):

$$\Delta E = \pi J \ln \frac{L}{a} \tag{5}$$

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- Crucial insight: Vortices are bound in pairs below critical temperature, and unbound above it
  - For $T < \frac{\pi J}{2k_B}$: $\Delta F > 0$ (bound vortex pairs)
  - For $T > \frac{\pi J}{2k_B}$: $\Delta F < 0$ (free vortices proliferate)

Renormalization Group Analysis

- Mapping to 2D Coulomb gas: vortices as charges with $\ln(r)$ interaction.
- Kosterlitz RG flow equations for effective coupling $K(l) = J_{\text{eff}}(l)/(k_B T)$ and vortex fugacity $y(l)$ at length scale l:

$$\frac{dK^{-1}(l)}{dl} = 4\pi^3 y(l)^2 + O(y(l)^4) \tag{8}$$

$$\frac{dy(l)}{dl} = (2 - \pi K(l))y(l) + O(y(l)^3) \tag{9}$$

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  - **Low-T** ($K > 2/\pi$): $y \to 0$ (vortices are irrelevant, QLRO prevails).
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- Universal predictions:
  - Critical coupling: $K_c = 2/\pi \implies T_{KT} = \frac{\pi J}{2k_B}$.
  - **Nelson-Kosterlitz jump**: Spin stiffness ρ_s (helicity modulus) jumps discontinuously at the transition:

$$\rho_s(T_{KT}^-) = \frac{2k_B T_{KT}}{\pi} \tag{10}$$

  - Correlation length above transition: $\xi(T) \sim \exp\left(b/\sqrt{T/T_{KT} - 1}\right)$

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  - Critical slowing down near phase transition
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  - Energy change for spin flip: $\Delta E = -J \sum_{\langle i,j \rangle} [\cos(\theta_i' - \theta_j) - \cos(\theta_i - \theta_j)]$
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  - Very inefficient near critical point due to large correlated regions
- More efficient approaches:
  - **Wolff cluster algorithm**: Updates entire clusters of spins
  - **Parallel tempering**: Exchanges configurations between different temperatures
  - Combination of both methods for optimal efficiency

Wolff Cluster Algorithm

- Efficiently updates entire clusters of correlated spins
  1. Choose a random reflection axis (unit vector $\mathbf{r}$)
  2. Pick a random initial site for the cluster
  3. Add neighboring spins with probability $P_{\text{add}} = 1 - \exp[\min(0, 2\beta J(\mathbf{S}_i \cdot \mathbf{r})(\mathbf{S}_j \cdot \mathbf{r}))]$
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- Our implementation:
  - Combines Wolff updates with parallel tempering
  - Optimally adapts the number of updates based on average cluster size
  - Uses numba JIT compilation for performance

Parallel Tempering

- Simulates multiple replicas at different temperatures $T_1 < T_2 < ... < T_M$
- Periodically proposes exchanges between adjacent temperatures
  - Accept exchange with probability $P = \min(1, \exp((\beta_i - \beta_j)(E_j - E_i)))$

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  - Configurations can diffuse across temperature space
- Implementation details:
  - Alternating even/odd exchanges for better mixing
  - Temperature spacing optimized for exchange acceptance rates
  - Combined with Wolff algorithm for optimal efficiency

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Our Implementation Approach

- Simulation structure:
  1. **Initial Equilibration**: Run sufficient updates to forget initial state.
  2. **Thermalization**: Ensure equilibrium is reached at each target temperature (often combined with equilibration).
  3. **Measurement**: Collect statistics over long runs in equilibrium.

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- Optimization techniques:
  - Numba ‘@njit‘ compilation for core simulation loops (Wolff update, energy calculation).
  - Vectorized operations with NumPy where possible.
  - Pre-computation of neighbor indices.
  - (Optional: Dynamic adjustment of updates per sweep based on avg. cluster size / acceptance rates).

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- Simulation workflow:
  - Define temperature range and system sizes L.
  - Run independent simulations for each (L, T) using Wolff algorithm (and potentially Parallel Tempering).
  - Thermalize, then collect measurements (Energy, Stiffness, etc.) over many sweeps.
  - Calculate averages and estimate errors (e.g., via Jackknife).
  - Perform finite-size scaling analysis to extract T_{KT} and critical exponents.

Key Observables

- Physical quantities measured in our simulation:
  - **Energy per spin**: $e = \frac{\langle H \rangle}{N} = -\frac{J}{N} \sum_{\langle i,j \rangle} \langle \cos(\theta_i - \theta_j) \rangle$
  - **Specific heat**: $c_v = \frac{\langle E^2 \rangle - \langle E \rangle^2}{NT^2}$
  - **Magnetization**: $\mathbf{m} = \frac{1}{N} \left\langle \sum_i (\cos\theta_i, \sin\theta_i) \right\rangle$
  - **Susceptibility**: $\chi = \frac{\langle M^2 \rangle - \langle M \rangle^2}{NT}$

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  - **Spin stiffness (Helicity modulus)**:

$$\rho_s = \langle H_x \rangle - \frac{\beta}{N} \langle I_x^2 \rangle \tag{11}$$

where $H_x = \sum_{\langle i,j \rangle_x} \cos(\theta_i - \theta_j)$ and $I_x = \sum_{\langle i,j \rangle_x} \sin(\theta_i - \theta_j)$

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  - **Vortex density**: $\omega_v = \frac{1}{N} \sum_{\text{plaquettes}} \left| \frac{1}{2\pi} \sum_{\text{edges}} \Delta\theta_{ij} \right|$
- Spin stiffness is crucial for locating T_{KT} through the universal jump

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  2. For each block b, compute estimate $\bar{O}_b$ by excluding block b
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- Benefits:
  - Correctly accounts for autocorrelations in MC data
  - Works well for derived quantities (non-linear functions)
  - More reliable than naive variance calculations
  - Essential for accurate error bars on quantities near transition

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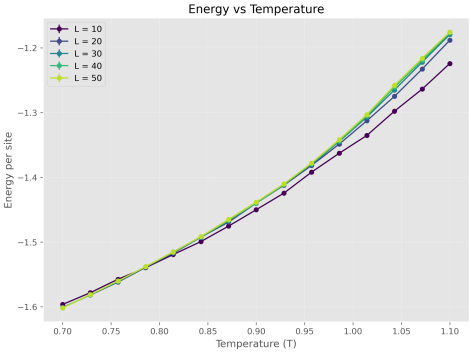
Energy and Specific Heat

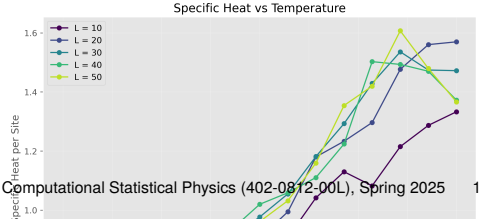
Internal Energy per Spin (e)

- Smoothly decreases with temperature.
- Weak dependence on system size L.
- No sharp features indicating the transition.

Specific Heat (c_v)

- Exhibits a broad peak around $T \approx 1.0J$.
- Peak height increases slightly with L.
- Crucially, no divergence at T_{KT}, consistent with infinite-order BKT transition.





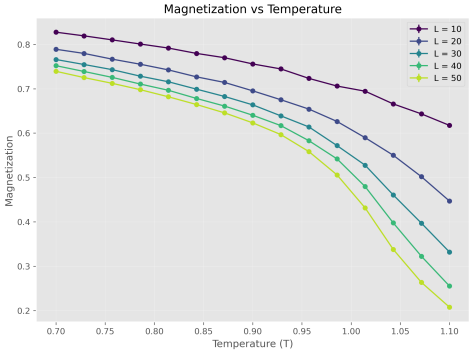
Magnetization and Susceptibility

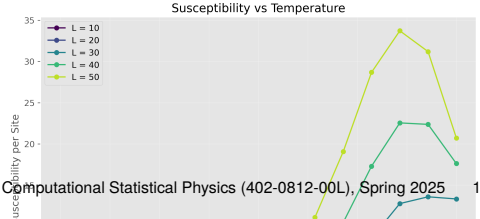
Magnetization ($|\mathbf{m}|$)

- Suppressed at finite T for finite L, consistent with Mermin-Wagner.
- Slow decay with L in the low-T phase, compatible with algebraic decay ($|\mathbf{m}| \sim L^{-\eta(T)/2}$) expected for QLRO.
- Faster decay in the high-T phase.

Susceptibility (χ)

- Develops a peak near the transition region.
- Peak height grows significantly with L.
- Diverges as $L \to \infty$ for $T \leq T_{KT}$.
- Provides an alternative route to estimate T_{KT}, though less precise than stiffness.





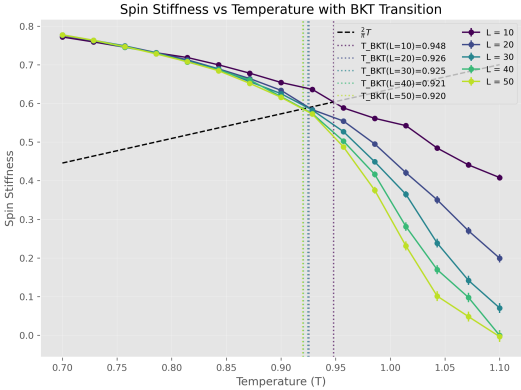
Spin Stiffness (Helicity Modulus ρ_s)

Identifying the BKT Transition

- Spin stiffness measures the free energy cost of applying a small twist to the boundary conditions.
- Decreases monotonically with temperature.
- Key indicator: Universal jump prediction by Nelson-Kosterlitz:

$$\rho_s(T_{KT}^-) = \frac{2k_B T_{KT}}{\pi}$$

- Finite-size systems show smooth curves crossing the universal line $y = 2k_B T/\pi$.
- Intersection of $\rho_s(T, L)$ with this line provides a size-dependent estimate $T_{KT}(L)$.
- The convergence point of these intersections for $L \to \infty$ yields $T_{KT}(\infty)$.
- Our data suggests $T_{KT} \approx 0.89 J/k_B$.

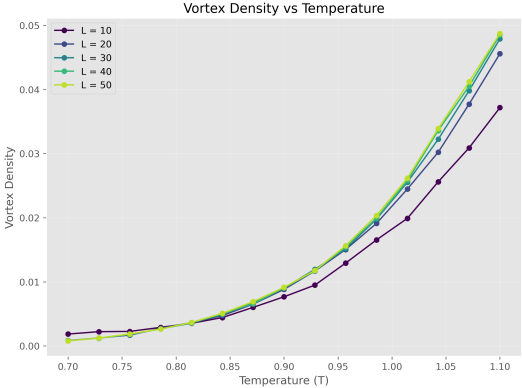


Plot shows ρ_s vs T for different L. The dashed line is $y = 2k_B T/\pi$.

Vortex Density

Role of Topological Defects

- Vortex density ω_v measures the concentration of topological defects.
- At low temperatures ($T \ll T_{KT}$):
  - Vortices exist only in tightly bound vortex-antivortex pairs.
  - Measured density is very low.
- Near T_{KT}:
  - Pairs start to unbind.
  - Density increases sharply.
- Above T_{KT}:
  - Proliferation of free vortices and antivortices.
  - Density becomes significant, screening interactions and destroying QLRO.
- Provides a microscopic view of the transition mechanism.



Plot shows average vortex density vs T for different L.

Finite-Size Scaling of $T_{KT}(L)$

Extracting the Transition Temperature

- Estimate $T_{KT}(L)$ for each size L from the intersection $\rho_s(T, L) = 2k_B T/\pi$.
- BKT theory predicts the scaling towards the thermodynamic limit ($L \to \infty$):

$$T_{KT}(L) = T_{KT}(\infty) + \frac{c}{(\ln(L/L_0))^2}$$

where c and L_0 are non-universal constants.

Finite-Size Scaling of $T_{KT}(L)$

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- Estimate $T_{KT}(L)$ for each size L from the intersection $\rho_s(T, L) = 2k_B T/\pi$.
- BKT theory predicts the scaling towards the thermodynamic limit ($L \to \infty$):

$$T_{KT}(L) = T_{KT}(\infty) + \frac{c}{(\ln(L/L_0))^2}$$

  where c and L_0 are non-universal constants.
- Plotting $T_{KT}(L)$ vs $1/(\ln L)^2$ should yield a linear relationship for large L.
- The y-intercept gives the estimate for $T_{KT}(\infty)$.

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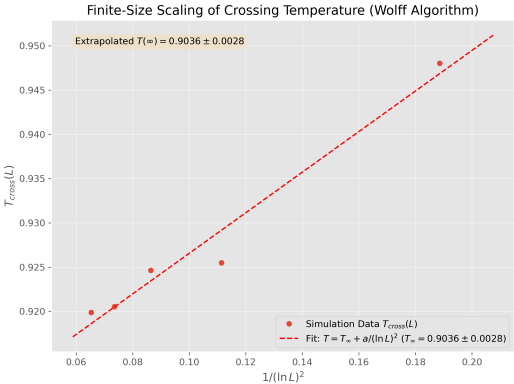
- The y-intercept gives the estimate for $T_{KT}(\infty)$.
- Our linear fit yields:

$$T_{KT}(\infty) \approx 0.893(4) J/k_B$$

- Result is consistent with established literature values (e.g., Ref. [1] and others cite $T_{KT} \approx 0.8929 J/k_B$).



Plot shows $T_{KT}(L)$ extracted from stiffness data vs $1/(\ln L)^2$. The line is a linear fit.

Outline

1. Introduction

2. Theoretical Background

3. Monte Carlo Methods

4. Implementation Details

5. Results and Analysis

6. Conclusion

Summary and Significance

- Successfully simulated the BKT transition in the 2D XY model:
  - Implemented efficient Wolff algorithm with parallel tempering
  - Measured key observables across multiple system sizes
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- Key findings:
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- Broader implications:
  - BKT transition represents a unique universality class
  - Demonstrates the power of topological defects in phase transitions
  - Provides insight into universal mechanisms across diverse physical systems
  - Highlights importance of advanced sampling techniques for critical phenomena

Future Directions

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  - Higher-dimensional models with topological transitions
- Applications:
  - Superconducting thin films and arrays
  - Quantum computing platforms based on 2D systems
  - Cold atom experiments with vortex dynamics
  - Theoretical connections to fundamental physics

References I

[1] Victor Drouin-Touchette. “The kosterlitz-thouless phase transition: an introduction for the intrepid student”. In: *arXiv preprint arXiv:2207.13748* (2022).

ETH zürich

Lara Turgut *lturgut@ethz.ch*
ETH Zurich, Department of Physics

Francesco Conoscenti *fconoscenti@ethz.ch*
ETH Zurich, Department of Information Technology and Electrical Engineering

Ben Bullinger *bben@ethz.ch*
ETH Zurich, Department of Computer Science